

Probability, Statistics & Data Analysis

1: Introduction, Descriptive Statistics and Visualisation

**Most people use statistics more like
a drunk person uses a lamp post;
more for support than illumination**

Andrew Lang

1: Introduction, Descriptive Statistics and Visualisation

What we will cover

- Course practicalities
- What are probability and statistics?
- Course overview
- Descriptive Statistics
 - Population vs Sample
 - Parametric vs Non-parametric
 - Mean, Median, Percentiles,
 - Median absolute deviation, standard deviation, variance
- Data visualisation
 - Dot plots, Histograms, Boxplots, Beeswarm plots, Violin plots





Course Practicalities

Where and when

- Tuesdays
 - except reading week, November 12th
- 12:00 – 14:00 RSM 131
 - Y3 UG BME and MBE
 - 12:00 to ~ 12:50, 13:00 to ~ 13:50
- 16:00 – 17:50 Huxley 213 Clore Lecture Theatre
 - BTV, iBSc, MSc
 - 16:00 to ~ 16:50, 17:00 to ~ 17:50
- Live in person
- Lectures recorded *for revision purposes* – can be watched on Panopto
- Please do attend in real time!
- I hypothesise that there is a correlation between attendance and marks
 - although I don't have evidence to support this, 'absence of evidence' is not 'evidence of absence'



Study Groups

- Some questions by hand, some on MATLAB
 - Bring laptops if you can
 - Can work in pairs/small groups
- Please attend your designated group (details should be on CELCAT)
- Format will be:
 - You try the material
 - During the study group, you ask for assistance from GTAs and peers
 - If a question comes up a lot, then the GTA may present to the group
 - Solutions will be uploaded after the study group

Wed	10:00	MSc Neurotech + HBR	301C
Fri	10:00	MSc Computational	303
	11:00	Y2 BTV + MSc Med Phys	228
Mon	12:00	MSc Biomaterials + Biomechanics	303
	14:00	Y3 UG A	301C
	14:00	Y3 UG B	303
	15:00	Y3 UG C	301C
	15:00	Y3 UG D	303
	16:00	Y3 UG E	303
	17:00	Y3 UG F	303

Coursework (20%)

- Widely requested
- MATLAB based coursework
- Practical statistical analysis of data
- Set on November 26th
- Due at 3pm on Friday December 13th

Exam (80%)

- 80-minute exam in summer term
- 2 questions with 4-8 sub-parts
- Sample exams on blackboard from reading week



Textbooks



- In general, I am yet to find a statistics textbook that I find consistently clear and helpful
- I have put together this course based on many sources with some of my own interpretations and explanations
- There is therefore not a single core textbook, but these will be good for extra reading

Freely available as Ebooks via the Leganto reading list

- Sheldon M. Ross: *Introduction to Probability and Statistics for Engineers and Scientists*
- Walpole, Myers, Myers, Ye: *Probability and Statistics for Engineers and Scientists*
- Andy Field, Jeremy Miles and Zoe Field: *Discovering Statistics using SPSS/R*
- Richard Johnson: *Miller & Freund's probability and statistics for engineers.*

Others that I recommend

- Jay L Devore: *Probability and Statistics for Engineering and the Sciences*

Getting Help

- Ask questions in lectures
 - Raise your hand
 - Pipe up!
- Study Groups
- Post questions on Ed
- Email – jvbsherwood@imperial.ac.uk
- The interwebs
 - <https://www.youtube.com/c/zedstatistics>
 - <https://www.youtube.com/c/ProfAndyField>

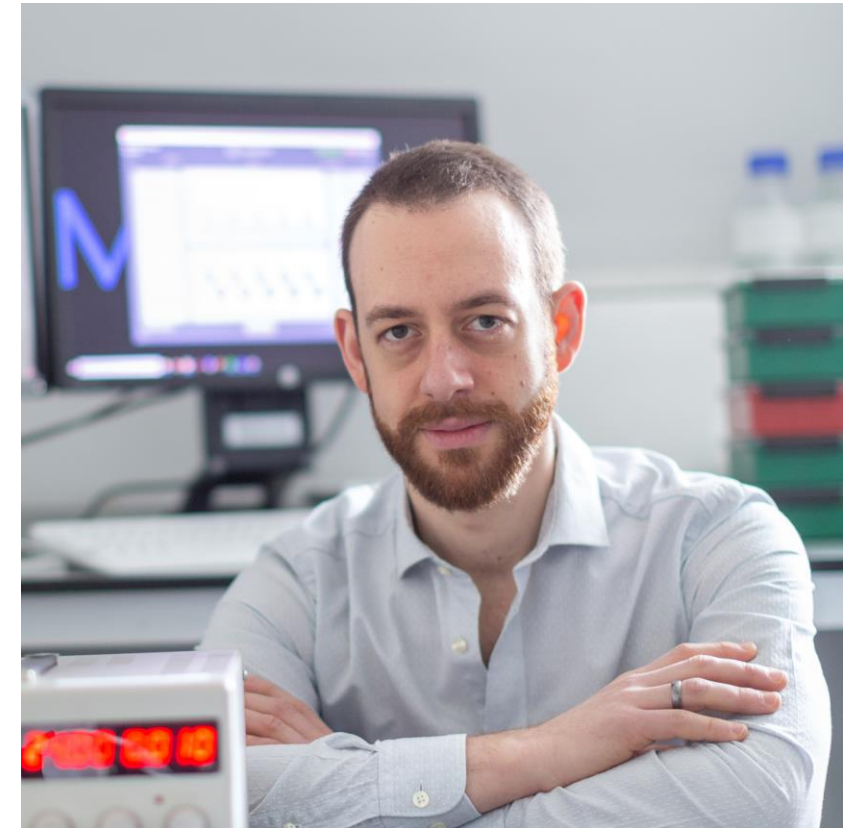


Dr Joseph van Batenburg-Sherwood

Or sometimes

Dr Sherwood
Dr Batenburg
Dr Battenburg
Dr Batenbrug
Dr van Bantenburg-Sherwood
Dr van Battenberg-Sherwood
Dr van Batenbourg-Sherwood
Dr van Batenburg
Dr von Botten-something
Prof ...
Van

(erroneously)



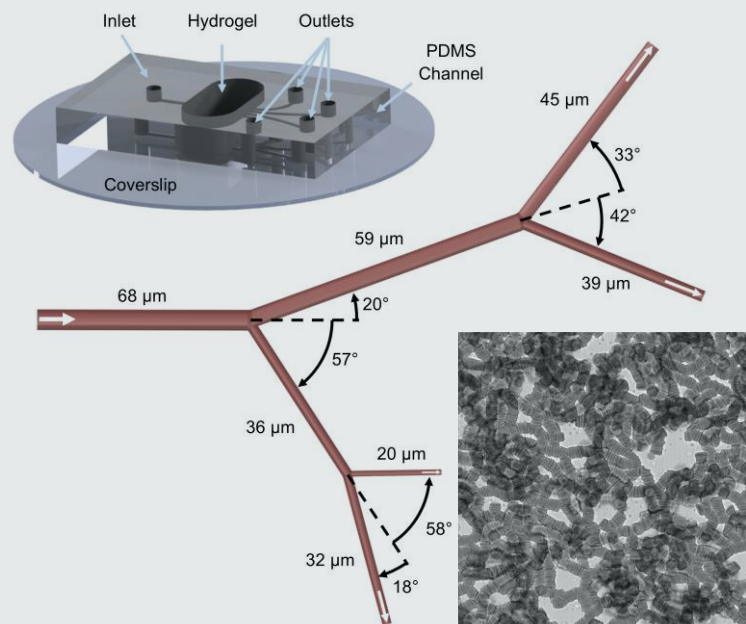
Who am I

and why am I here?

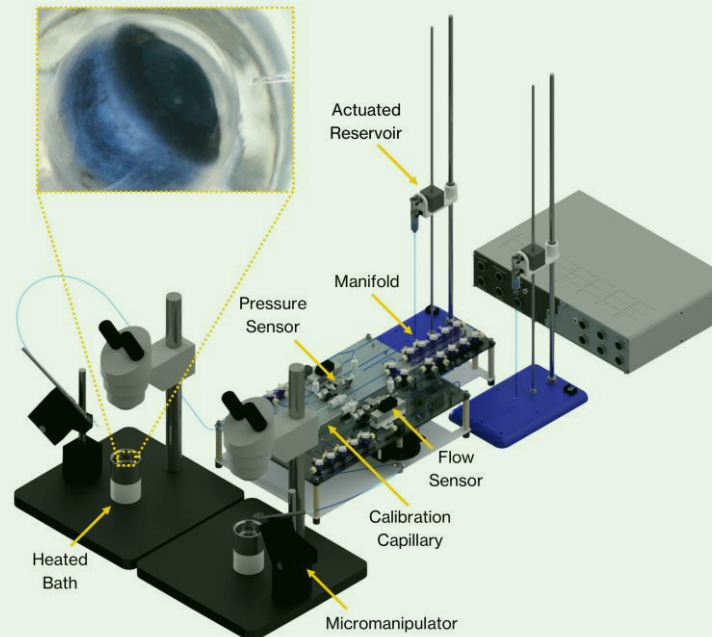
My Research

- Mechanical Engineering background (no statistics!)
- Biofluid Mechanics (learned to love statistics)
- Specialise in experimental approaches, combined with simulations and theory
- **Solving problems** and **addressing unanswered questions** related to biological flows

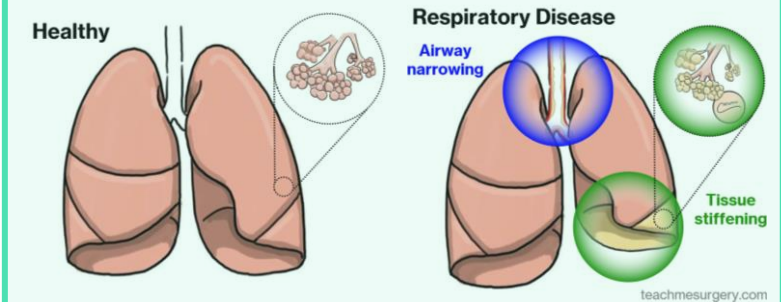
Blood



Eyes



Lungs





Probability and Statistics

What are they?



Example - WBC count in chemotherapy

- Chemotherapy reduces white blood cell (WBC) count as a side-effect
- A new drug is proposed (a colony stimulating factor) that promises to reduce this side-effect
- Does it work or not? Or more correctly 'how well does it work'?
- Clinical trial
 - Test some patients with the current medicine (controls)
 - Test some patients with the new medicine (treated)
 - Quantify number of WBC from blood tests

Controls	2,800	3,700	8,200	5,700	4,900	...
Treated	7,400	6,300	3,800	5,300	3,300	...

- How many patients do we need to test to get an answer?
- How much can we trust the answer?

WBC count – motivation for statistics



- We need to do some analysis to get from the data to a conclusion

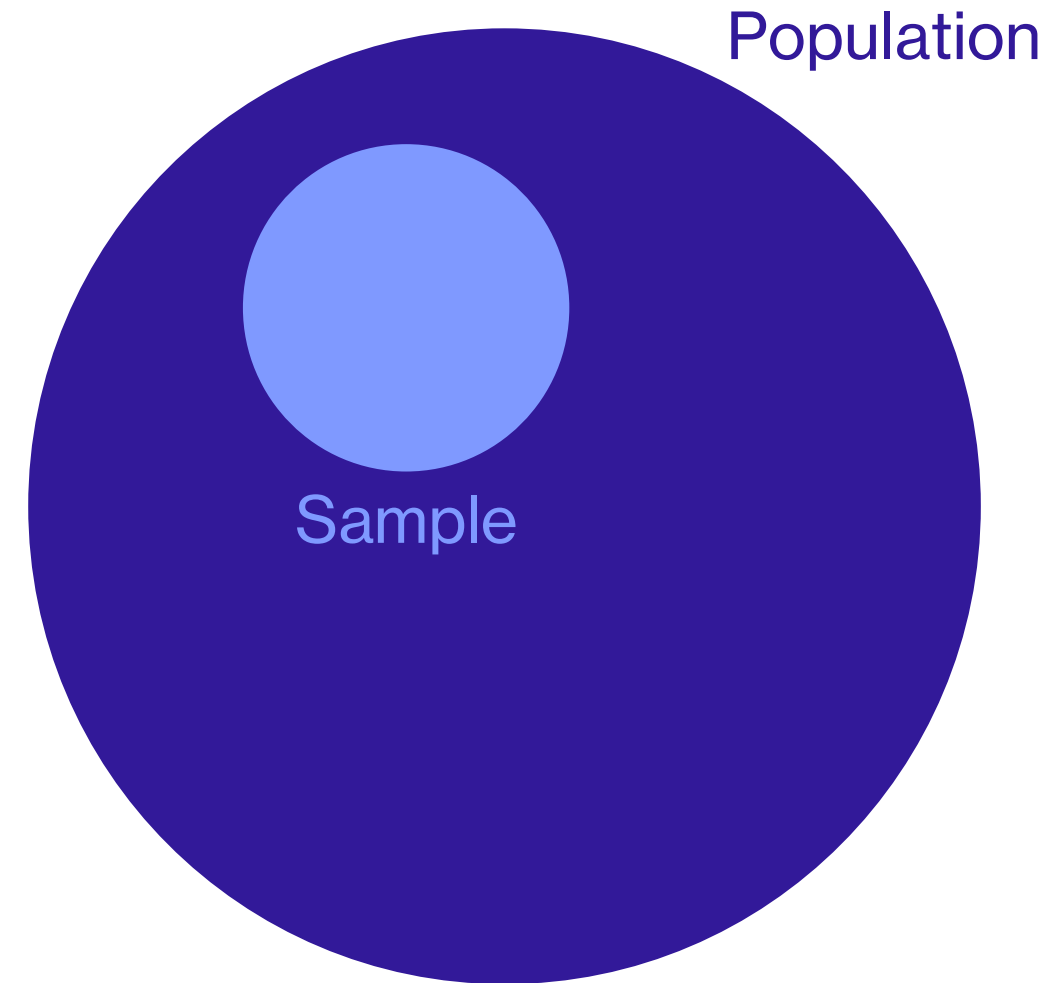
Without the analysis, the data is meaningless

- E.g. we conclude that the drug works
 - If we're right
 - Improve patient outcomes
 - Save lives
 - If we're wrong
 - Make patient outcomes no better and waste time and money (that could have gone elsewhere)
 - Make patient outcomes worse
- Hence, without the best possible analysis, we risk making no improvement or making things worse
- Here we will learn how to minimise the chances of this!

How well does the drug work?



- Cannot test everyone with cancer – ‘population’
- Instead must test a subset – ‘sample’
- Use the outcome to **infer** (deduce, conclude, informed-guess) something about the population
- Statistics and probability provide ways to communicate between the sample and population



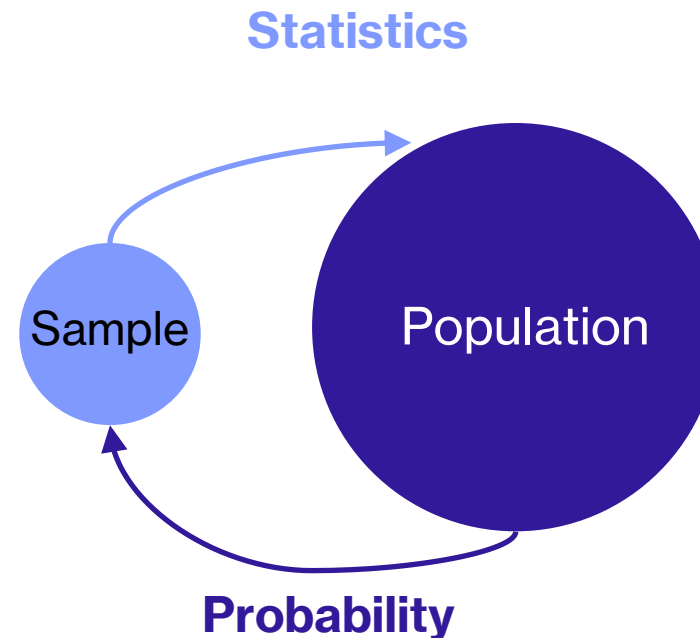
Probability and Statistics

Descriptive Statistics

- Describe the key characteristics of a data set (sample or population)
- Graphs, measures of spread and location

Inferential Statistics

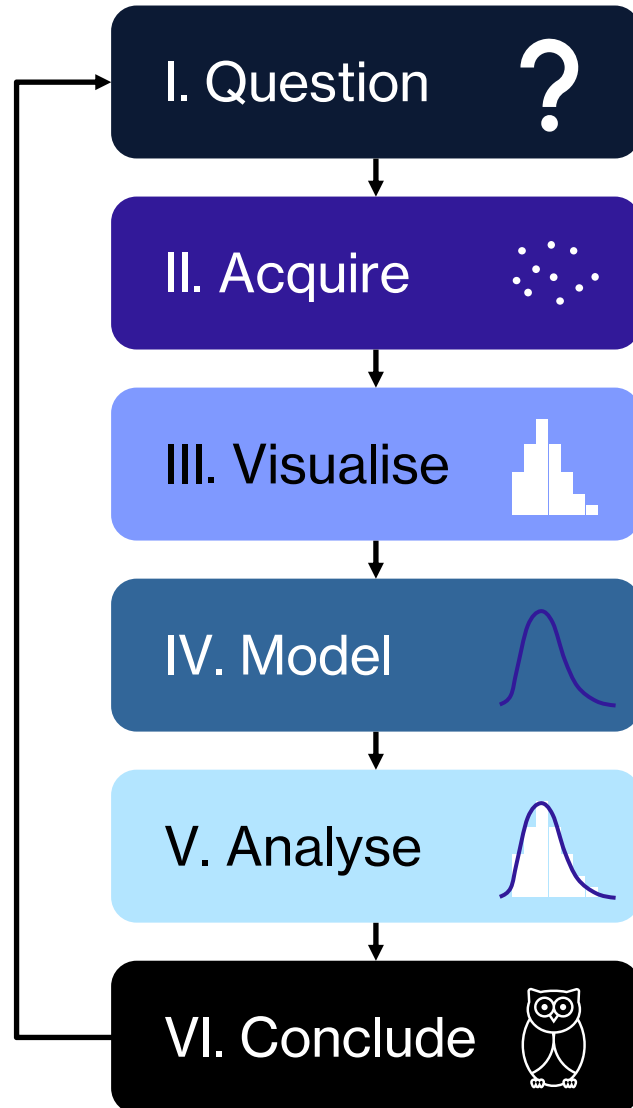
- We use a sample to estimate something about the population
- Examples
 - Does a drug being trialled increase WBC count? If so, how?
 - Does a particular prosthetic improve comfort and dexterity
 - Are mice from a population with a specific gene knockout more susceptible to diabetes



Probability

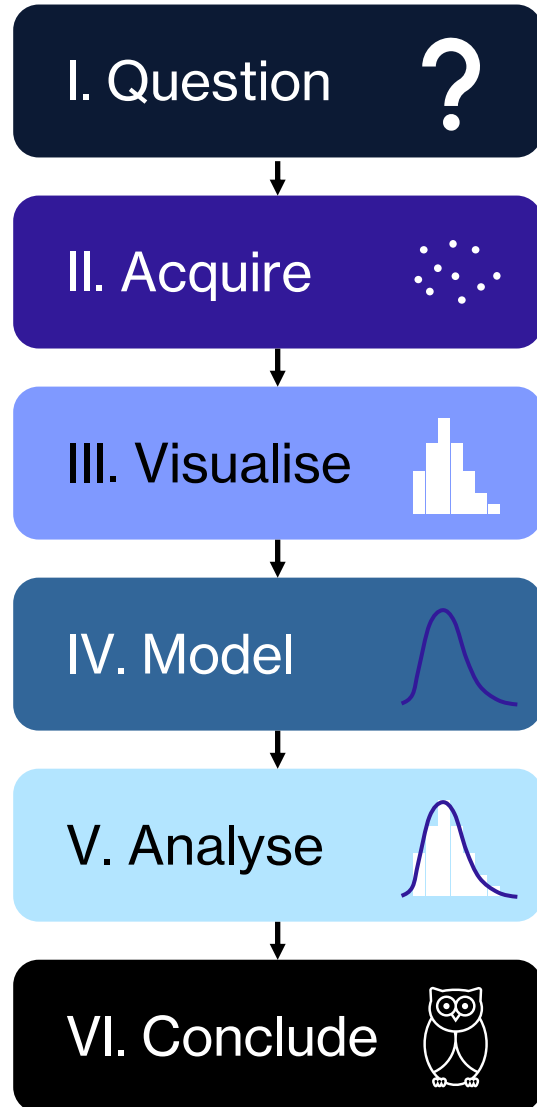
- We already 'know' the details of the population
- We want to calculate how likely a certain outcome is for a given sample
- Examples
 - How likely is it that a group of 40 people contains only healthy people if we know 1% of the population have a disease
 - Mostly, we use probability to understand the foundations of statistics

A Framework for Data Analysis



- I. Question
 - start by considering the question you want to ask
- II. Acquire Data
 - collect a sample of data relevant to your problem
- III. Visualise
 - plotting the data gives you intuition
- IV. Model
 - propose a model for the data (population behaviour) and check if it is valid
- V. Analyse
 - if valid, use the model. If not, use an approach that does not need a model
- VI. Conclude
 - report the results of your analysis, with due respect paid to uncertainty

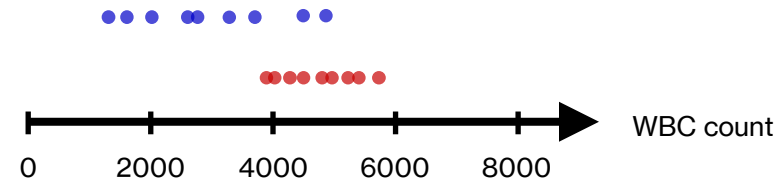
Framework Example – WBC count



- I. Does the new drug improve patient outcomes?
II. Data

Controls	2,800	3,700	8,200	5,700	4,900	...
Treated	7,400	6,300	3,800	5,300	3,300	...

- III. Visualise:



- IV. Model: The data are sampled from a normal distribution
– Check if the data are consistent with this model
- V. (Assuming they are) estimate the average change in WBC count, uncertainty on that estimate and variability in the change in WBC count
- VI. Consider whether
- the average change in WBC count is practically significant
 - the average change in WBC count is statistically different from zero
 - the variability in the WBC counts is important



Probability and Statistics

They are tools to help us with Data Analysis

Probability and Statistics for Bioengineering

- You are not mathematicians, you are engineers
- Engineering is essentially problem solving
- The problem here is: how best to
 - Ask my question?
 - Design my experiment
 - Evaluate my data
- Analogy – torque screwdriver
 - Used to apply only a predefined torque when tightening screws to avoid shearing the screw
 - We need to know
 - Why we would use this instead of e.g. a normal screwdriver, a chisel or a hammer
 - We don't need to know
 - The exact mechanism of the screwdriver
 - Although it might improve our ability to use it properly
- There are as many statistical tests as there are different tools
- I aim to provide you with a basic statistics 'toolkit'
- I hope to help you enjoy data analysis (or at least dislike it less)



Proof vs Practice

- Analytical proof
 - Often depends on tricks and algebra
 - Realm of the mathematician
- Practice
 - Engineering is a lot about intuition
 - Can be gained through practice
 - Computers can easily simulate data (or we can import real data sets)
 - Numerical ‘proofs’ (sorry mathematicians)
 - Shows we understand the concept properly
 - Also, real data sets normally large enough to warrant use of a computer

- To that end, we will use MATLAB in this course
- The coursework will be MATLAB based
- See Introduction to MATLAB course from Tom Ouldridge on Blackboard Page if required

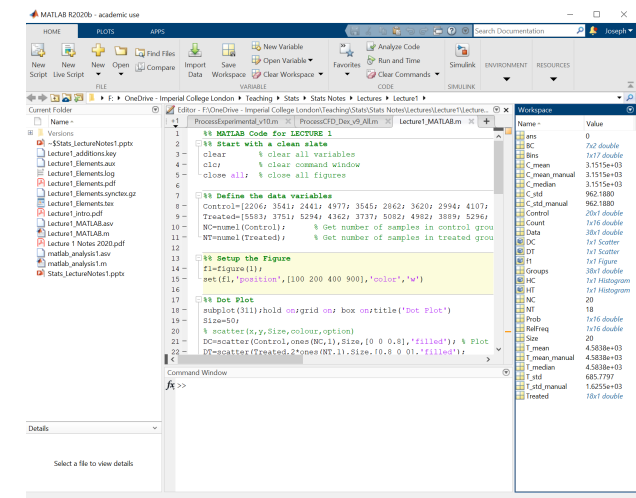
$$\begin{aligned} \sum_{i=1}^n (x_i - \bar{x})^2 &= \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 \\ &= \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\ &= \sum_{i=1}^n x_i^2 - n\bar{x}^2 \end{aligned}$$

SC

$$\begin{aligned}\mathbb{E}\left(\sum_{i=1}^n [(x_i - \mu) - (\bar{x} - \mu)]^2\right) &= \mathbb{E}\left(\left(\sum_{i=1}^n (x_i - \mu)^2\right) - n(\bar{x} - \mu)^2\right) \\ &= \left(\sum_{i=1}^n \mathbb{E}((x_i - \mu)^2)\right) - n\mathbb{E}((\bar{x} - \mu)^2) \\ &= \left(\sum_{i=1}^n \text{Var}(x_i)\right) - n\text{Var}(\bar{x})\end{aligned}$$

and by definition,

$$\begin{aligned} \mathbb{E}(s^2) &= \mathbb{E}\left(\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}\right) \\ &= \frac{1}{n-1} \mathbb{E}\left(\sum_{i=1}^n [(x_i - \mu) - (\bar{x} - \mu)]^2\right) \\ &= \frac{1}{n-1} \left[\left(\sum_{i=1}^n \text{Var}(x_i)\right) - n \text{Var}(\bar{x}) \right] \end{aligned}$$





Course Overview

What we're going to cover

Course Structure

1. **Course Introduction, Descriptive Statistics and Visualisation**
 2. **Probability Theory**
 3. **Discrete Random Variables**
 4. **Continuous and Joint Random Variables**
 5. **The Normal Distribution, Statistics and Intervals** Reading Week
 6. **Null-hypothesis statistical testing and z-tests**
 7. **T-tests and Assumptions**
 8. **Non-parametric tests and Regression**
 9. **ANOVAe**
- } Coursework

Reference Sheet and Statistical Tables

Purple boxes mean it's on the reference sheet

- Provided for the exam
 - More than just a formula sheet
 - I regularly use as a quick reference
- Custom statistical tables
 - For use in the exam
 - Practice using during the course

PROBABILITY, STATISTICS AND DATA ANALYSIS

REFERENCE SHEET 2024/2025

Question Acquire Visualise Model Analyse Conclude

?

Descriptive Statistics

Mean		Variance		Standard deviation	
Population $\mu = \frac{1}{N} \sum_{i=1}^N x_i$	Sample $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Population $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$	Sample $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$	Population $\sigma = \sqrt{\sigma^2}$	Sample $s = \sqrt{s^2}$
Percentiles $[n(1-p)]$ values $\geq \eta(p)$		Median Odd n : $\tilde{x} = x_{(\frac{n+1}{2})}$ Even n : $\tilde{x} = 0.5(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)})$		Median absolute deviation $MAD = k x_{(\frac{n+1}{2})} - \bar{x} $ $k = 1.4826$ if underlying distribution is normal	

Probability

Frequentist Probability $P(A) = n(A)/n$	Product rule $n = n_1 \times n_2 \times \dots \times n_k$	Permutations $P_{k,n} = \frac{n!}{(n-k)!}$	Combinations $C_{k,n} = \binom{n}{k} = \frac{n!}{k!(n-k)!}$
Intersection and Union $P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Conditional probability $P(A B) = \frac{P(A \cap B)}{P(B)}$	Independence If $P(A B) = P(A)$ or $P(A \cap B) = P(A)P(B)$	Bayes Theorem $P(A_i B) = \frac{P(B A_i)P(A_i)}{\sum_{j=1}^k P(B A_j)P(A_j)}$

Random Variables

Expected Values

Discrete $E(X) = \mu_X = \sum_{i=1}^n x_i p(x_i)$	Continuous $E(X) = \mu_X = \int_{-\infty}^{\infty} x f(x) dx$	Joint discrete $E[h(X,Y)] = \sum_{i=1}^n \sum_{j=1}^m h(x_i, y_j) p(x_i, y_j)$	Joint continuous $E[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) f(x,y) dx dy$
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Variance

Discrete $V(X) = \sigma_X^2 = \sum_{i=1}^n (x_i - E(X))^2 p(x_i)$	Continuous $V(X) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx$
--	--

Marginal PMF/PDF

Discrete $p_X(x) = \sum_{j=1}^m p(x, y_j)$ $p_Y(y) = \sum_{i=1}^n p(x_i, y)$	Continuous $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$
--	--

Independence

X and Y are independent if for every pair x, y

Discrete $p(x, y) = p_X(x)p_Y(y)$	Continuous $f(x, y) = f_X(x)f_Y(y)$
--------------------------------------	--

Covariance

Discrete $cov(X, Y) = \sum_{i=1}^n \sum_{j=1}^m (x_i - E(X))(y_j - E(Y))p(x_i, y_j)$	Continuous $cov(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - E(X))(y - E(Y))f(x, y) dx dy$
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Cumulative Distribution Functions

Discrete $F(x) = P(X \leq x) = \sum_{x_i \leq x} p(x_i)$	Continuous $F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$
---	---

Other

Percentiles $p = P[\eta(p)]$ $= \int_{-\infty}^{\eta(p)} f(y) dy$	Population Correlation coefficient $\rho = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$
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Key Distributions and Random Variables

Bernoulli $p(x; \alpha) = \begin{cases} 1 - \alpha & \text{if } x = 0 \\ \alpha & \text{if } x = 1 \end{cases}$ $E(X) = p, V(X) = pq$	Binomial $X \sim Bin(n, p)$ $b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$ $E(X) = np, V(X) = npq$	Poisson $X \sim Pois(\lambda)$ $\lambda = np, p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$ $E(X) = \lambda, V(X) = \lambda$
Normal $X \sim N(\mu, \sigma^2)$ $E(X) = \mu_X, V(X) = \sigma_X^2$	Standard Normal $Z = (X - \mu_X)/\sigma_X, Z \sim N(0, 1)$ $E(Z) = 0, V(Z) = 1$	Sample mean (CLT) $\bar{X} \sim N(\mu, \sigma^2/n)$ $E(\bar{X}) = \mu_X, V(\bar{X}) = \sigma_X^2/n$

PROBABILITY, STATISTICS AND DATA ANALYSIS

STATISTICAL TABLES 2024/25

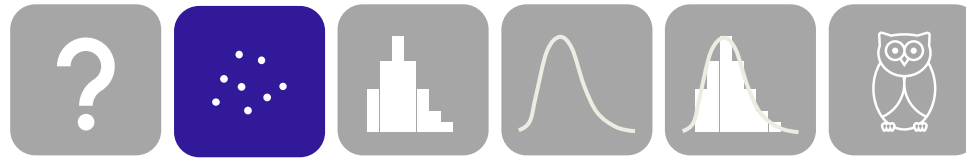
Table 1: Cumulative Distribution Function for Standard Normal Distribution, $\Phi(z)$

z (dp 2)

$\Phi(z)$

Positive values only given. For negative values, use $\Phi(-z) = 1 - \Phi(z)$
Note e.g. $z_{1-\alpha/2}$ is equivalent to $\Phi(z) = 1 - \alpha/2$ or equivalently $z = \Phi^{-1}(1 - \alpha/2)$

Second decimal place of z										
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9978	0.9979	0.9980	0.9981	0.9982	0.9983
2.9	0.9984	0.9985	0.9986	0.9987	0.9988	0.9989	0.9990	0.9991	0.9992	0.9993
3.0	0.9994	0.9995	0.9996	0.9997	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999
3.1	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.2	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.3	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.4	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999

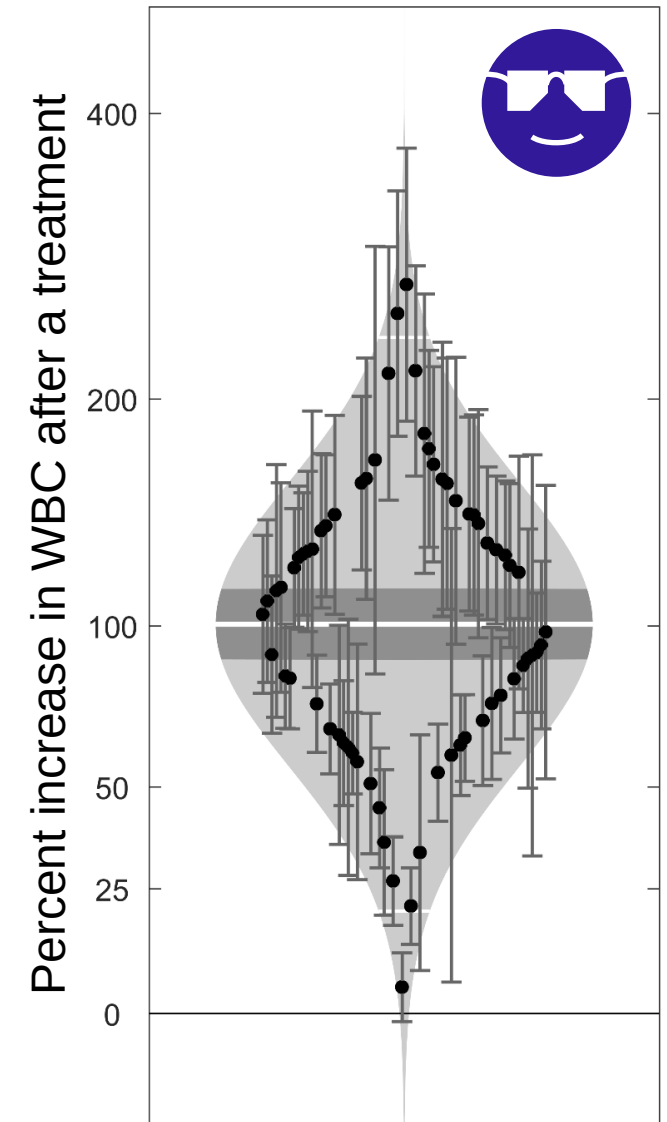
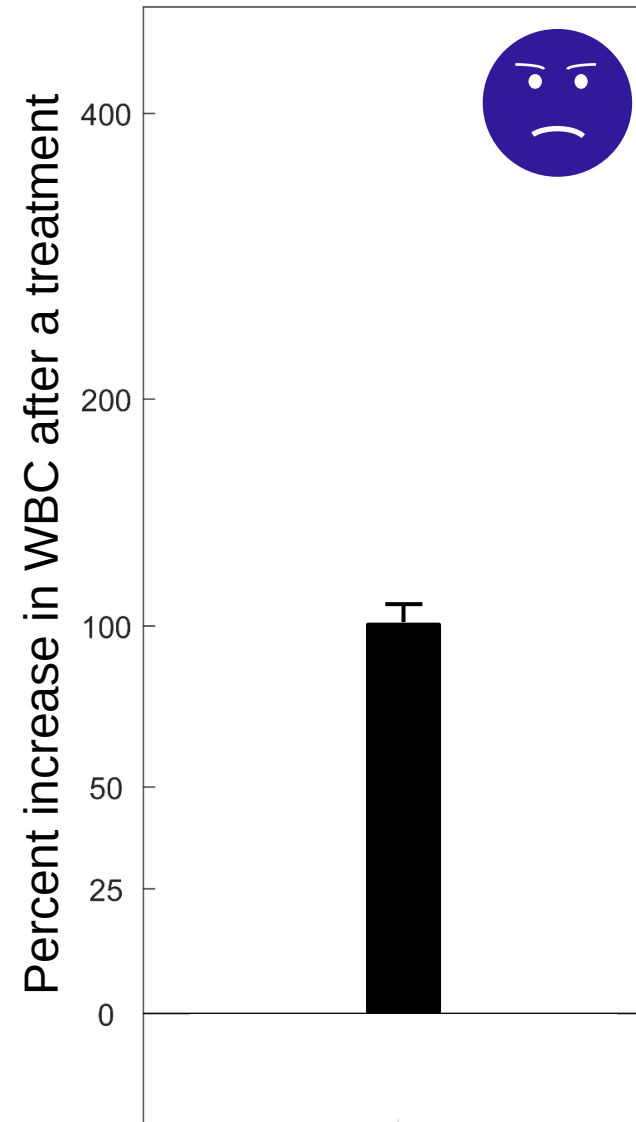


Data Visualisation (with MATLAB)

Communicating the data

Data Visualisation

- The aim of data visualisation is to communicate the data as effectively as possible
- 'A picture is worth a thousand words'
- There is no single 'correct' way to visualise data, just as there is no 'correct' way to form a sentence
- However, just as there are poor ways to form sentences
- There are poor ways to visualise data (e.g. bar charts)

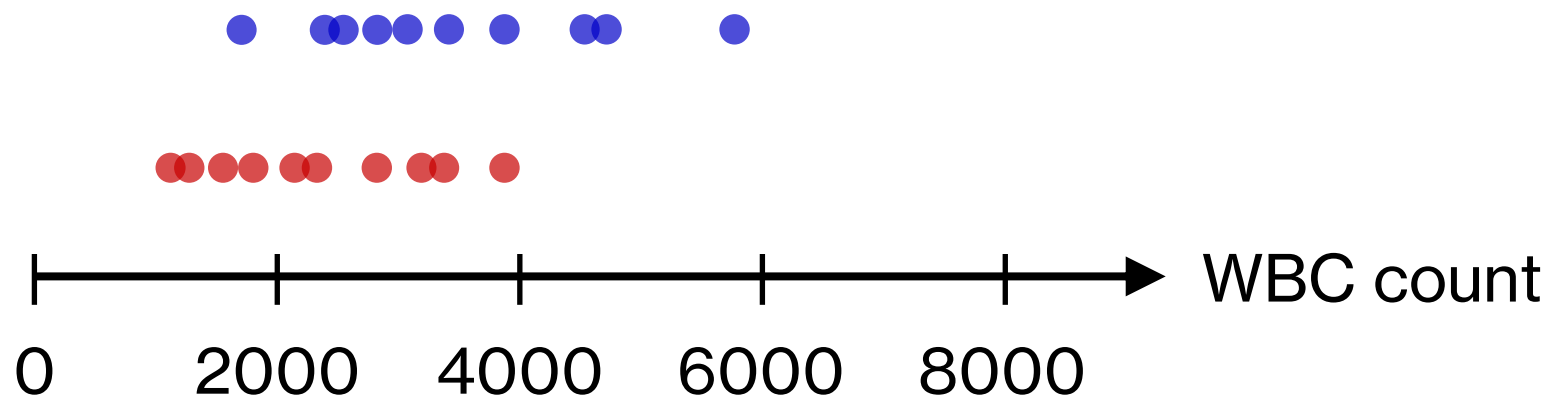


Dot Plots

Hint/Rule 1

Wherever possible, show all data points

- Each observation is represented by a dot
- Can be used horizontally or vertically
- Can be used in conjunction with other visualisation methods



Hint/Rule 2

Plot data points as points, not lines!



- We will use MATLAB for data analysis and visualisation
- In class examples, I will use screenshots
- Full code provided after lecture for you to try for yourself
 - Great way to test and develop understanding and intuition
 - Practical experience for processing your own data
 - Powerful and flexible graphing
 - MATLAB central – code sharing site
- Anything Excel can do, MATLAB can do better
- Statistics-specific packages exist, *R*, SPSS, GraphPad etc.
 - But none can do data processing and visualisation as well (including Python)

Purple bar



MATLAB Overview



The screenshot shows the MATLAB R2020b interface. The top toolbar includes options like 'New Script', 'New Live Script', 'Find Files', 'Import Data', 'Save Workspace', 'New Variable', 'Open Variable', 'Analyze Code', 'Run and Time', 'Clear Commands', 'Simulink', 'ENVIRONMENT', and 'RESOURCES'. The 'Current Folder' pane on the left lists files like 'Versions', '\$Stats_LectureNotes1.pptx', 'Lecture1_additions.key', 'Lecture1_Elements.aux', 'Lecture1_Elements.log', 'Lecture1_Elements.pdf', 'Lecture1_Elements.synctex.gz', 'Lecture1_intro.pdf', 'Lecture1_MATLAB.asv', 'Lecture1_MATLAB.m', 'Lecture 1 Notes 2020.pdf', 'matlab_analysis1.asv', 'matlab_analysis1.m', and 'Stats_LectureNotes1.pptx'. The 'Editor' pane in the center shows a MATLAB script for 'LECTURE 1' with sections for clearing variables, defining data, and plotting. The 'Workspace' pane on the right lists variables such as 'ans', 'BC', 'Bins', 'C_mean', 'C_mean_manual', 'C_median', 'C_std', 'C_std_manual', 'Control', 'Count', 'Data', 'DC', 'DT', 'f1', 'Groups', 'HC', 'HT', 'NC', 'NT', 'Prob', 'RelFreq', 'Size', 'T_mean', 'T_mean_manual', 'T_median', 'T_std', 'T_std_manual', and 'Treated', each with its size and data type. The 'Command Window' at the bottom shows the prompt 'fx >>'. Annotations include: 'Current Folder Path' pointing to the top toolbar; 'Variables details: size and type' pointing to the 'Workspace' pane; 'Variables' pointing to the list of variables in the 'Workspace' pane; 'Current Folder Contents' pointing to the 'Current Folder' pane; '.m file or 'script' editor' pointing to the 'Editor' pane; 'Command window' pointing to the 'Command Window' pane; and 'Workspace' pointing to the 'Workspace' pane.

Current Folder Path

Variables details:
size and type

Variables

Current Folder Contents

.m file or
'script' editor

Command window

Workspace

MATLAB Scripts or .m files



%% creates sections

Collapses sections

To turn on
Preferences>Editor/Debugger>Code folding> Enable "Sections"

; suppresses output from appearing in command window

Create 'handle' f1

Call 'handle' f1

Choose a property

Set the property values

% creates comment

Creates a clean slate

```
%% MATLAB Code for LECTURE 1
%% Start with a clean slate
clear      % clear all variables
clc;       % clear command window
close all; % close all figures

%% Define the data variables
Control=[2206; 3541; 2441; 4977; 3545; 2862; 3620;
Treated=[5583; 3751; 5294; 4362; 3737; 5082; 4982;
NC=numel(Control);      % Get number of samples in
NT=numel(Treated);      % Get number of samples in

%% Setup the Figure
f1=figure(1);
set(f1,'position',[100 200 400 900],'color','w')
```

A handle is a reference to an object: a name

Data Importing and help

Enter data manually

```
%% Define the data variables
```

```
Control=[2206; 3541; 2441; 4977; 3545; 2862
```

```
Treated=[5583; 3751; 5294; 4362; 3737; 5082
```

```
save('Lecture1_Data','Control','Treated')
```

```
clear
```

```
load('Lecture1_Data')
```

```
NC=numel(Control);
```

```
% Get number of sam
```

```
NT=numel(Treated);
```

```
% Get number of sam
```

Save workspace or specific variables to .mat file

Load workspace or specific variables from .mat file

numel
an in-built MATLAB function

```
>> help numel
```

numel Number of elements in an array or subscripted array expression.

N = **numel**(A) returns the number of elements, N, in array A, equivalent to **PROD**(**SIZE**(A)).

matlab numel



Images

Examples

Vs length

Videos

In Python

News

Books

Maps

Flights

About 327,000 results (0.44 seconds)



MathWorks

<https://www.mathworks.com> > ... > Matrices and Arrays

Number of array elements - MATLAB numel

n = **numel**(A) returns the number of elements, n, in array A, equivalent to **prod**(**size**(A)).

Examples. collapse all. Number of Elements in 3-D Matrix.

Setting Up Figures



Position and size on screen (left bottom width height)

Sets borders to white

Figure handle

```
%% Setup the Figure
f1=figure(1);
set(f1,'position',[100 200 400 900],'color','w')
```

311
312
313

221	222
223	224

```
%% Dot Plot
subplot(311);hold on;grid on; box on;title('Dot Plot')
```

Creates a subplot
(Rows, Columns, Number)

If number of subplots is
greater than 9, use commas.

E.g.

`subplot(4,4,11)`

Multiple plots on the
same axes

Background
grid

Plot edges

Figure title.
Avoid use for
reports etc.

Creating a Dot Plot

...and making it pretty

R G B R G B

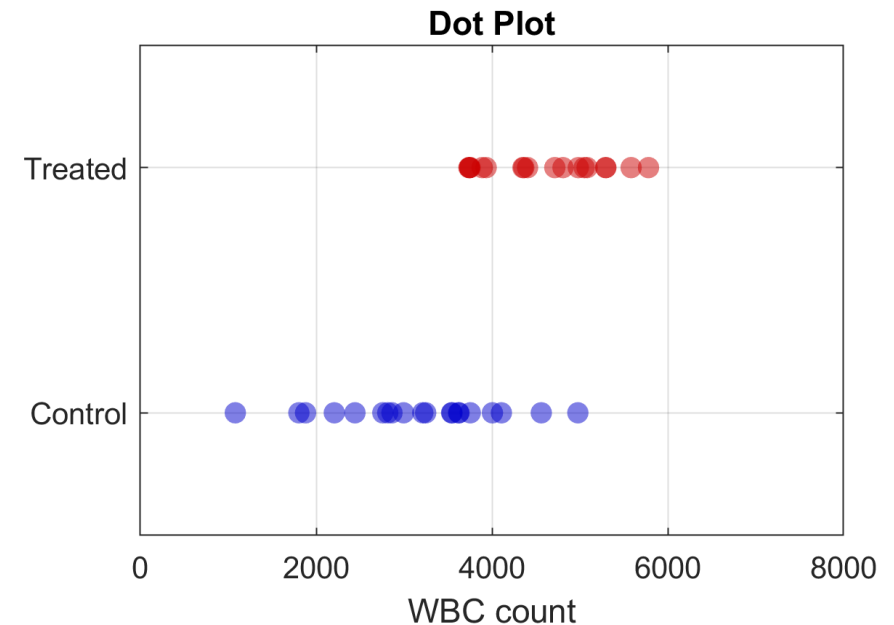
```
Col={[0 0 0.8];[0.8 0 0]}; % Create a cell array with two RGB colour values
```

% Dot Plot

```
subplot(151);hold on;grid on; box on;title('Dot plot')
Size=50; % This will determine how big the markers are
% scatter(x,y,Size,colour,option)
DC=scatter(Control,ones(NC,1),Size,Col{1},'filled'); % Plot
DT=scatter(Treated,2*ones(NT,1),Size,Col{2},'filled');

alpha(DC,0.5) % Make markers transparent
alpha(DT,0.5)

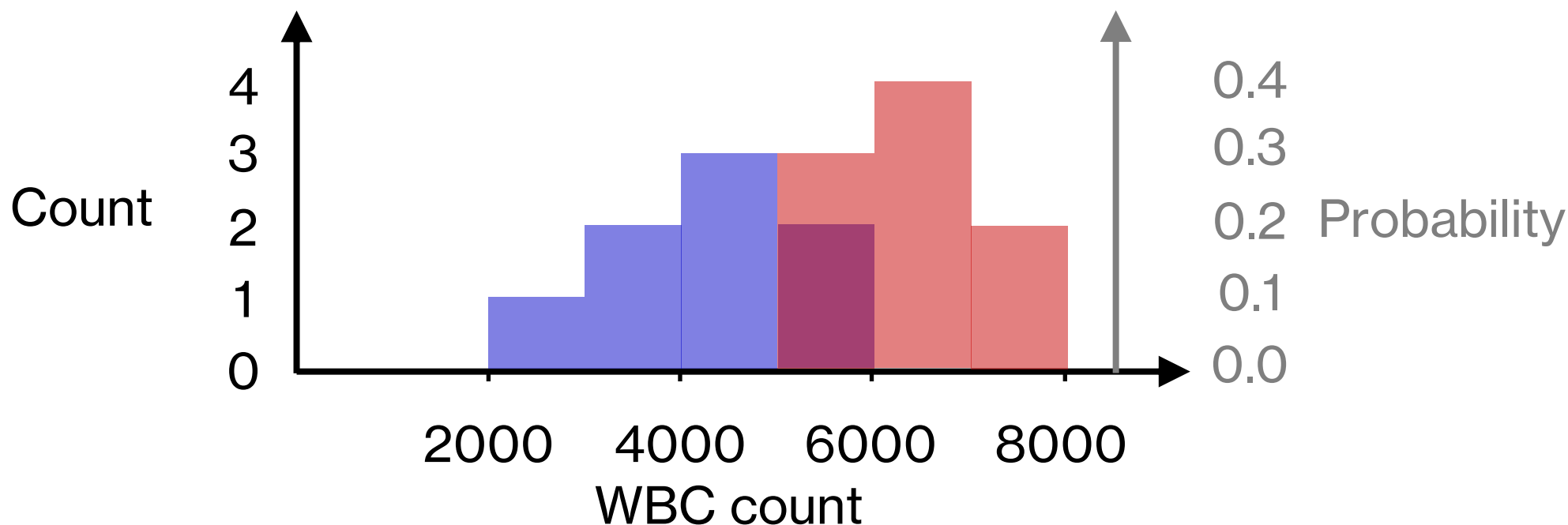
axis([0 8000 0.5 2.5]) % Set axis limits
set(gca,'ytick',[1 2],'yticklabel',{'Control','Treated'});
xlabel('WBC count') % set x axis label
```



Hint/Rule 3
Transparency is key (pun intended)

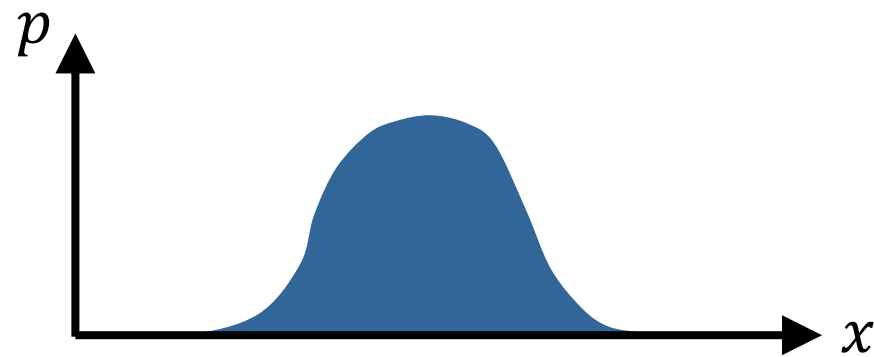
Histograms

- Count how often the data fall in a certain range of values
- Subdivide the x axis into suitable class intervals (typically of equal width) – ‘bins’
- Choose bin width to best represent the data
- Optional: compute the relative frequency (‘Probability’)

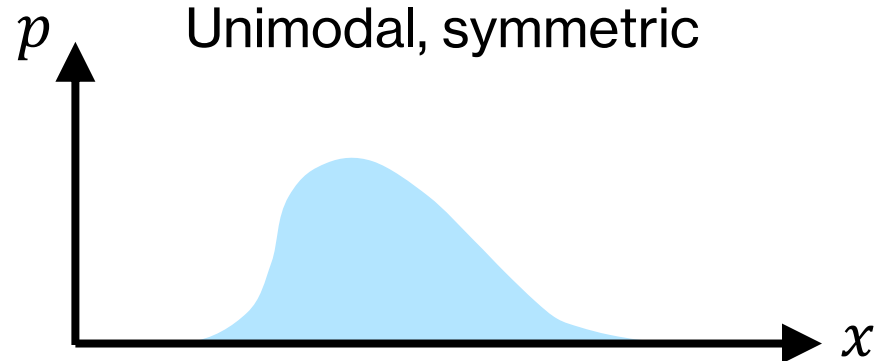


Probability Distribution Shapes

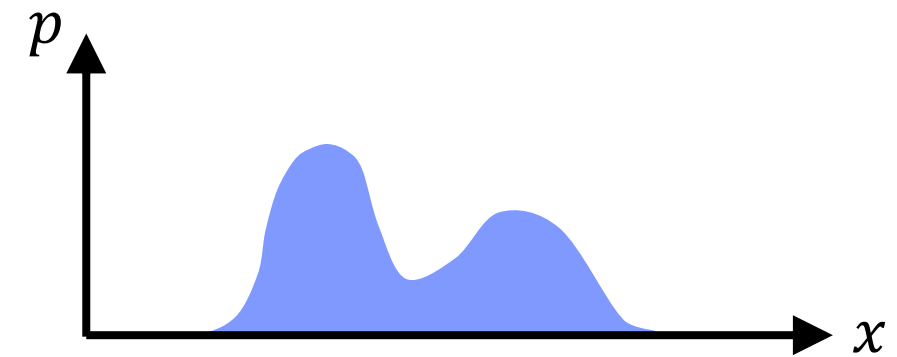
- Histograms are a way to visualise Probability Distributions
- This can help inform us what model to use



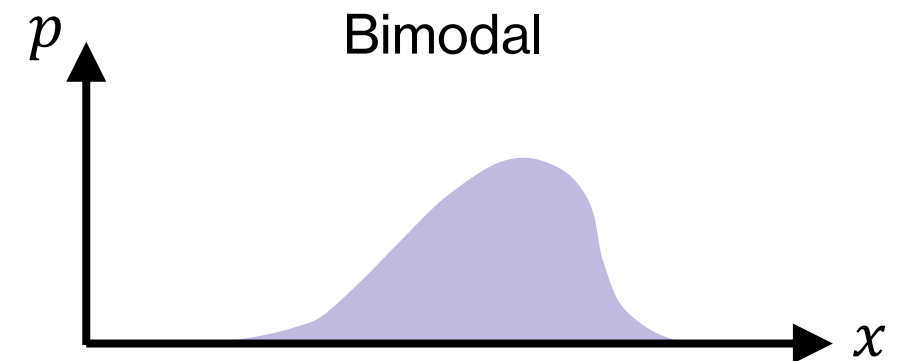
Unimodal, symmetric



Positively Skewed



Bimodal



Negatively Skewed

Creating a Histogram

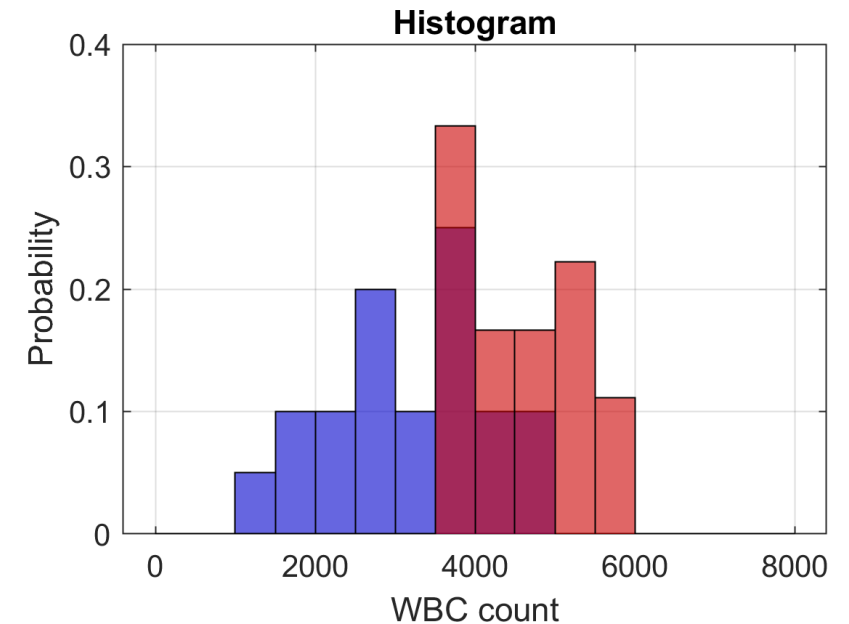
min: interval: max

```
% Histogram
subplot(152); hold on; grid on; box on; title('Histogram')
Bins=0:500:8000; % number of bins
HC=histogram(Control,Bins); % Create histogram with predefined
Count=HC.Values; % Extract the count for each bin
HT=histogram(Treated,Bins);

set(HC,'facecolor',Col{1},'normalization','probability')
set(HT,'facecolor',Col{2},'normalization','probability')

ylabel('Probability') % set y axis label
xlabel('WBC count') % set x axis label
yticks(0:0.1:0.3) % sets the yticks

% check relative frequency
Prob=HC.Values; % Export the probabilities from the histogram
RelFreq=Count/sum(Count); % Calcualte probabilities manually
sum(RelFreq-Prob); % Should be zero if it worked
```



Note: when we changed the normalisation,
we changed the values in HC.Values
This is why we saved the Count values

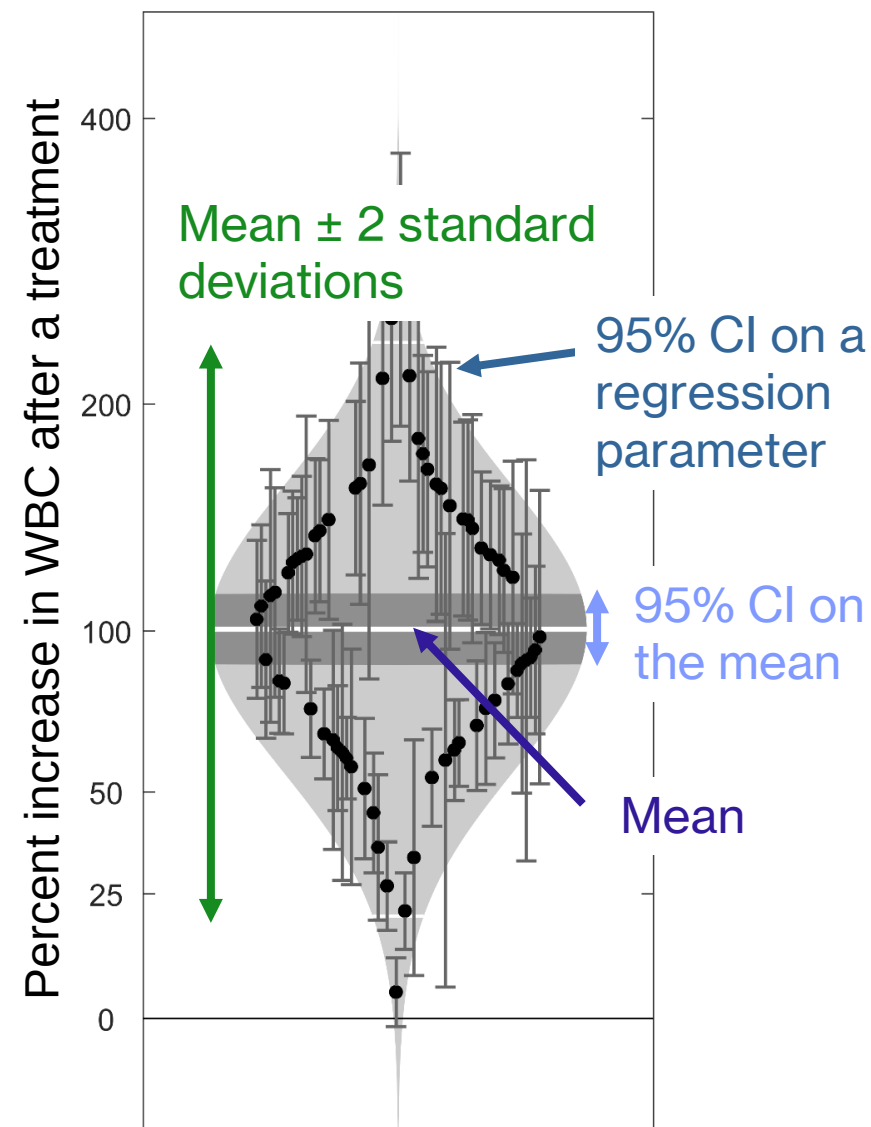


Descriptive Statistics

Measures of location and spread

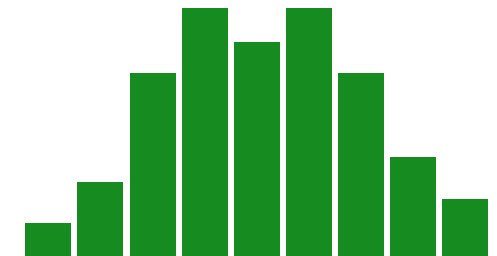
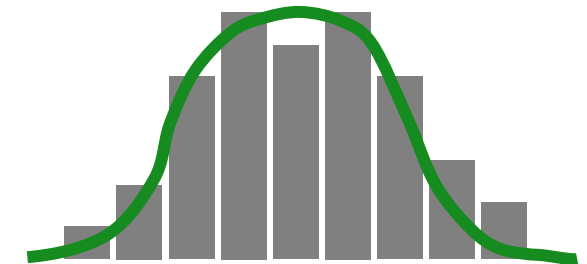
Descriptive Statistics

- When we have a lot of data and want to compare to a reference, or e.g. to a different treatment
- Descriptive statistics allow us to communicate the key features of a dataset efficiently
- Correct choice of descriptors is important
- For measures of location (averages), it could be
 - Arithmetic mean, geometric mean, median, trimmed mean....
- For 'measures of spread' (variability)
 - Variance, standard deviation, median absolute deviation or multiples thereof, interquartile range
- From which we can calculate
 - Margin of error, confidence intervals, prediction intervals
- By the end of the course, you will know what these all mean and which to use



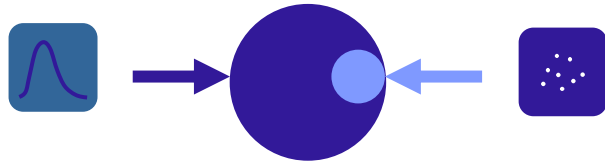
Parametric vs Non-Parametric Statistics

- Parametric statistics and procedures
 - Application or interpretation relies on some assumption about the underlying probability distribution – a model!
 - Generally more ‘powerful’ – lower uncertainty for a given sample size
 - Must be coupled with tests to confirm assumptions (this is done a lot less often than it should be)
- Non-parametric statistics and procedures
 - Application or interpretation does not rely on assumptions about the underlying probability distribution
 - Generally less ‘powerful’ – higher uncertainty for a given sample size
 - Generally more robust – not as sensitive to errors in assumptions or outliers
- More details to follow when we consider hypothesis testing
- For measures of location and spread, if I say e.g. standard deviation is parametric, I mean that it’s interpretation typically involves reference to an underlying probability distribution (in this case it is a parameter of the normal distribution)



Population vs Sample Values

- Sample Mean, Sample standard deviation, Sample Median etc. are only estimates of the true population values



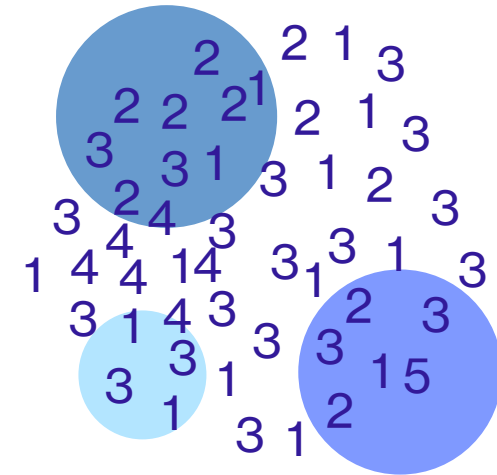
- These estimates have uncertainty associated with them and thus we need statistics to evaluate them
- First, we will look into understanding the behaviour of the population values, using probability

Population: $N = 50, \mu = 2.4$

Sample 1: $n = 10, \bar{x} = 2.2$

Sample 2: $n = 4, \bar{x} = 2.0$

Sample 3: $n = 6, \bar{x} = 2.7$





Measures of Location

Identifying where data are: averages

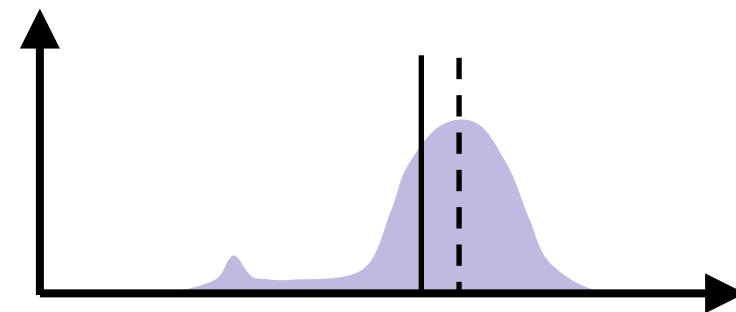
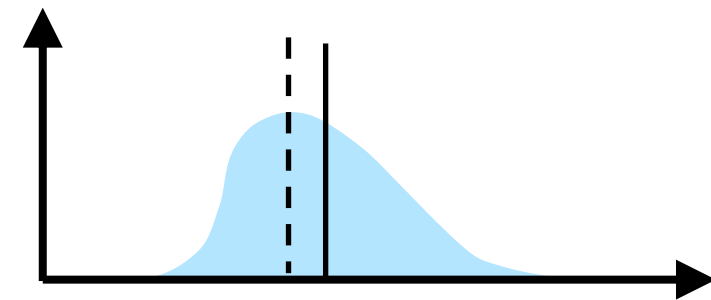
Measures of location

- Used to describe roughly 'where' data are
- An 'average'
- Typically Mean or Median
 - Mode isn't much use
- The key is how uncertain we are about our 'average'
- We'll get into this later in the course

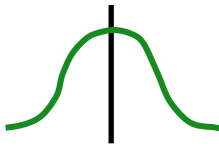
Hint/Rule 4

Always report the mean with the confidence interval on it (once you know how to)

————— Mean
----- Median



Sample Mean



The sample mean (arithmetic) from n observations is

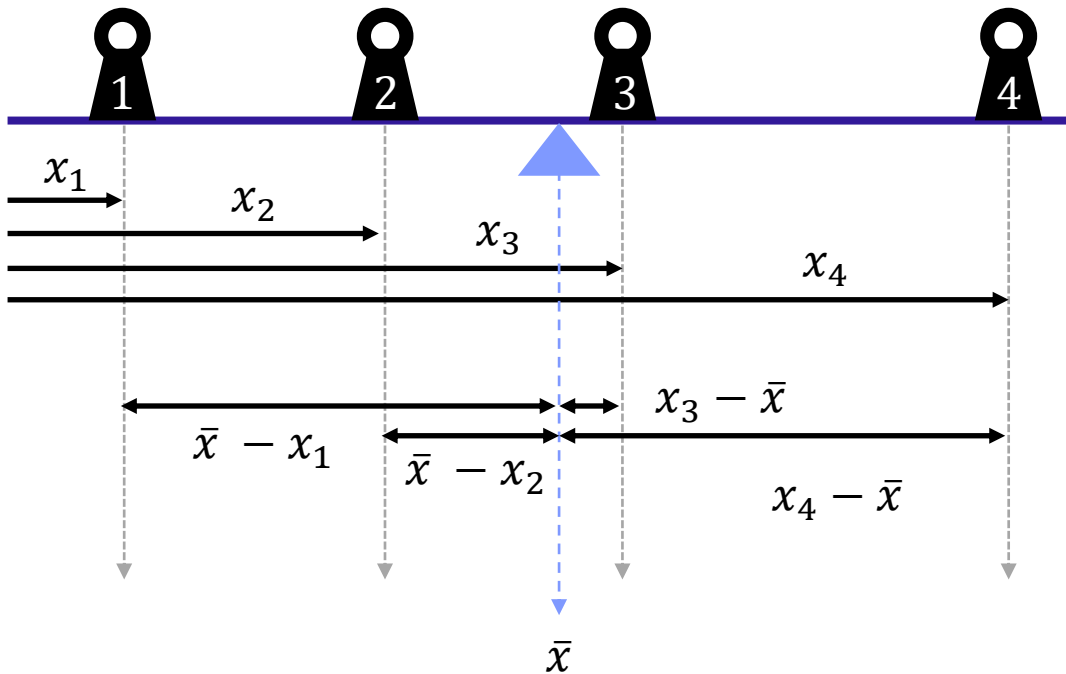
$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

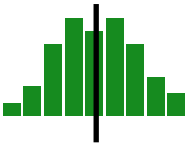
The sum of the deviations from the mean is zero

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} = \bar{x}n - \bar{x}n = 0$$

$$\bar{x} = \frac{x_1 + x_2 + x_3 + x_4}{4}$$

$$(\bar{x} - x_1) + (\bar{x} - x_2) + (\bar{x} - x_3) + (\bar{x} - x_4) = 0$$





Sample Median

- A non-parametric measure of location
- The middle value of the data when ordered from smallest to largest
- Obtained by ordering the n observations from smallest to largest (`sort`), with repeated values included so that every sample observation appears in the ordered list

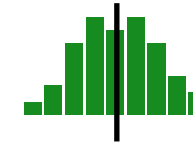
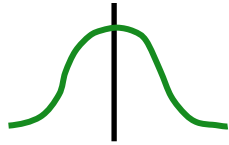
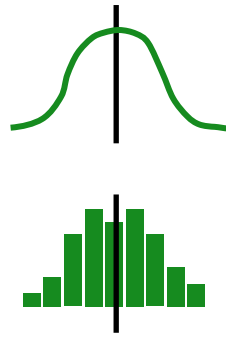
$$\tilde{x} = \begin{cases} \text{If } n \text{ is odd: the single middle value,} \\ x\left(\frac{n+1}{2}\right) \\ \text{If } n \text{ is even: the mean of the two middle values} \\ 0.5 \left(x\left(\frac{n}{2}\right) + x\left(\frac{n}{2} + 1\right) \right) \end{cases}$$

- More robust to 'outliers' than the mean

Find the Mean and Median

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\tilde{x} = \begin{array}{ll} \text{odd: } & x\left(\frac{n+1}{2}\right) \\ \text{even: } & 0.5 \left(x\left(\frac{n}{2}\right) + x\left(\frac{n}{2} + 1\right) \right) \end{array}$$



$$x_i = [1000, 367, 194, 13, 238, 436]$$

mean (x) , median (x)

Blue bar for
examples

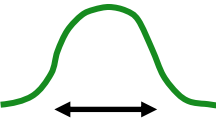




Measures of Spread

Identifying how variable data are

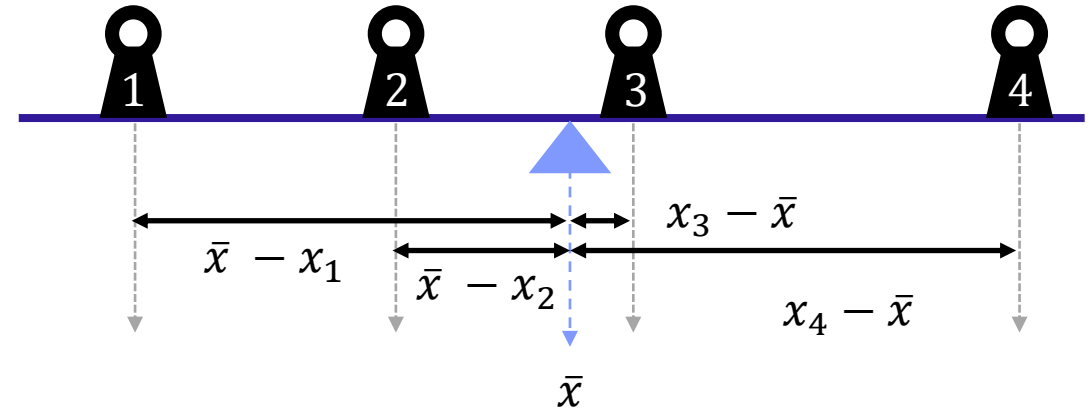
Sample Variance



- On average, how far are data points from the mean?
- Square solves sum equalling zero, but gives variance in x^2 units

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Reason for $n - 1$ addressed later when we talk about bias
- Sum squares notation

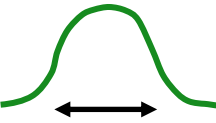


$$\frac{(\bar{x} - x_1) + (\bar{x} - x_2) + (\bar{x} - x_3) + (\bar{x} - x_4)}{n} = 0$$

$$s^2 = \frac{(\bar{x} - x_1)^2 + (\bar{x} - x_2)^2 + (\bar{x} - x_3)^2 + (\bar{x} - x_4)^2}{4 - 1}$$

$$s^2 = \frac{S_{xx}}{n-1} \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

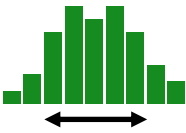
Sample Standard deviation



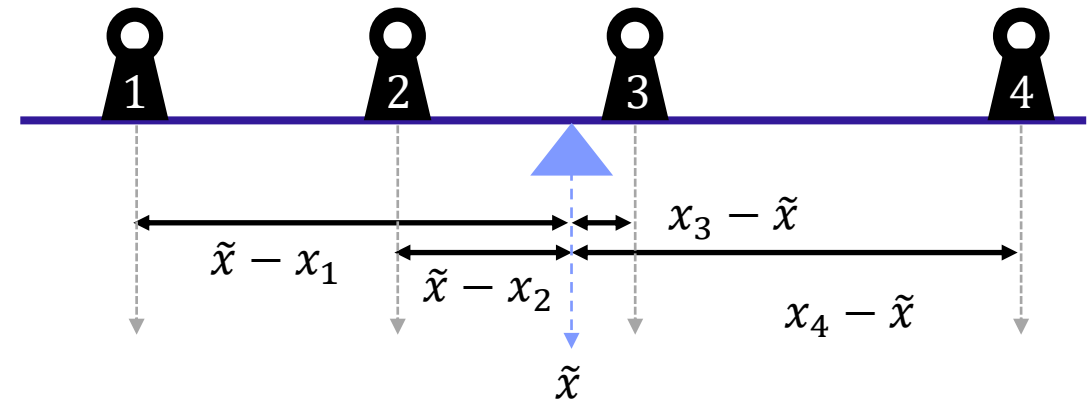
- Variance has value in certain calculations (see Normal Distributions later), but has unhelpful units
- The standard deviation has the same units as x so is better as a measure of uncertainty
- Mean $\pm$ Multiples of standard deviations contain useful proportions of the data, e.g. ~95% of data are in the range $\bar{x} \pm 2s$
- Almost a Root Mean Square Deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Sample Median Absolute Deviation



- A non-parametric measure of spread
- As an alternative to taking the square of each deviation from the mean
- Take the absolute value of each deviation from the median, then take the median



$$MAD = k|\widetilde{x_i - \tilde{x}}|$$

- The factor k scales for equivalence to standard deviation, typically 1.4826

Find the MAD and SD

$$x_i = [1, 2, 6]$$

$$MAD = k|\widetilde{x_i - \tilde{x}}|$$

$$k = 1.4826$$

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

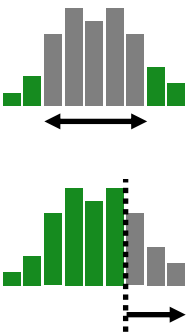
Note, MATLAB doesn't do the k , and requires argument 1 or will do mean absolute deviation from the mean

→ `k*mad(x,1) , std(x)`

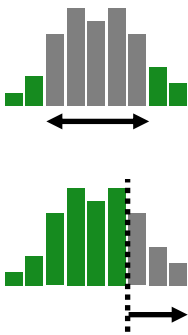
Sample Percentiles (discrete)

- Useful for describing proportions of a data set relative to a given number
- E.g. 200 people take an exam, and the 75th percentile is 72
- If an individual gets more than 72, they
 - Are ‘in the 75th percentile’ or ‘in the upper quartile’
 - Got higher marks than at least 75% of the class
- In general for a data set (specific definition we choose)

$100(1 - p)\%$ of values are greater than or equal to the $100p^{th}$ percentile $\eta(p)$



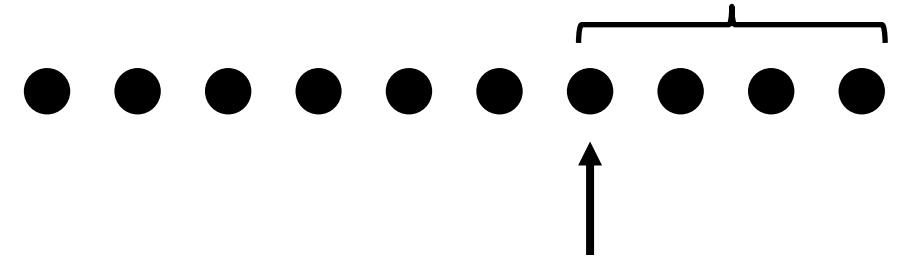
Sample Percentiles (discrete)



$100(1 - p)\%$ of values are greater than or equal to the $100p^{th}$ percentile $\eta(p)$

- Example – find the 60th percentile η of $X=[1,2,3,4,5,6,7,8,9,10]$
 - $p = 0.6$
 - 40% of the values should be larger than η
 - We could choose any value in the range

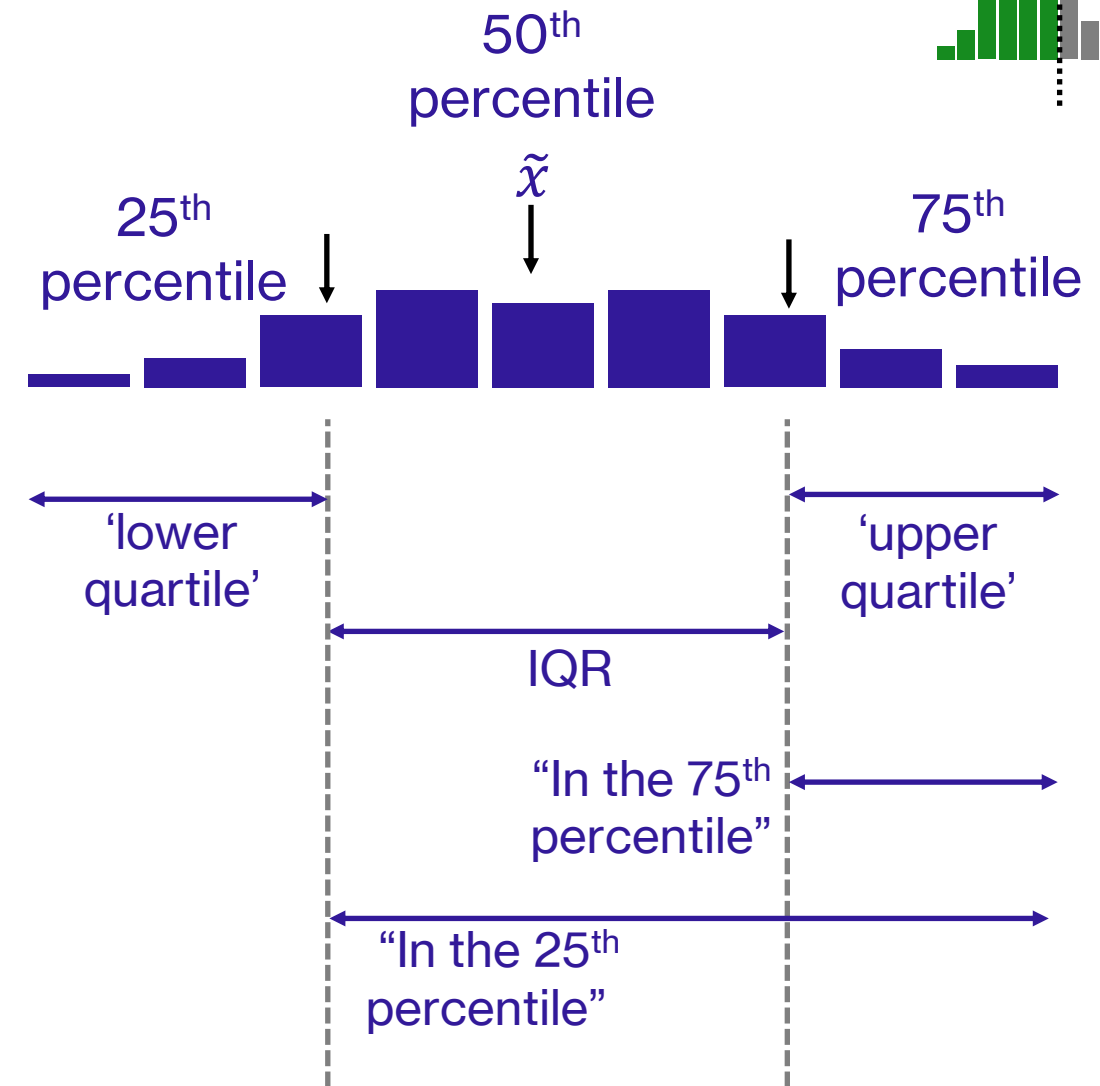
$$6 < \eta \leq 7$$
- In this course, when calculating manually, we select $\eta = 7$
 - i.e. a specific value from the data set
 - Calculate using $\lceil n(1 - p) \rceil$ values $\geq \eta(p)$
- Computer algorithms will interpolate, in this case to give $\eta = 6.5$



$$\lceil n(1 - p) \rceil \text{ values } \geq \eta(p)$$

Sample Percentiles Definitions

- Sample median is 50th percentile
- Below 25th percentile – ‘lower quartile’
- Above 75th percentile – ‘upper quartile’
- Between 25th and 75th quartile is the interquartile range (IQR)
- General terminology ‘*in* the 100 p^{th} percentile’ means ‘**above** $\eta(p)$ ’



Locate the IQR

$\lceil n(1 - p) \rceil$ values $\geq \eta(p)$



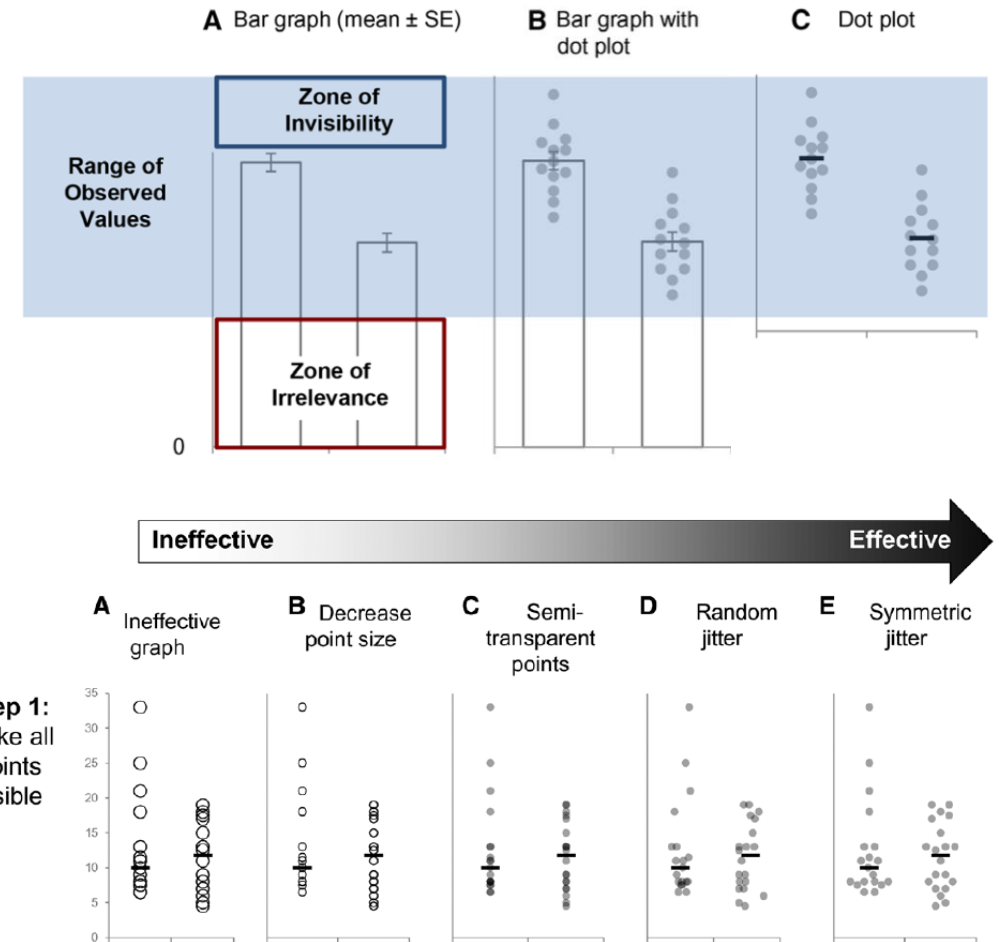


Sample Visualisation

Boxplots and Beeswarm plots

Why not use bar charts?

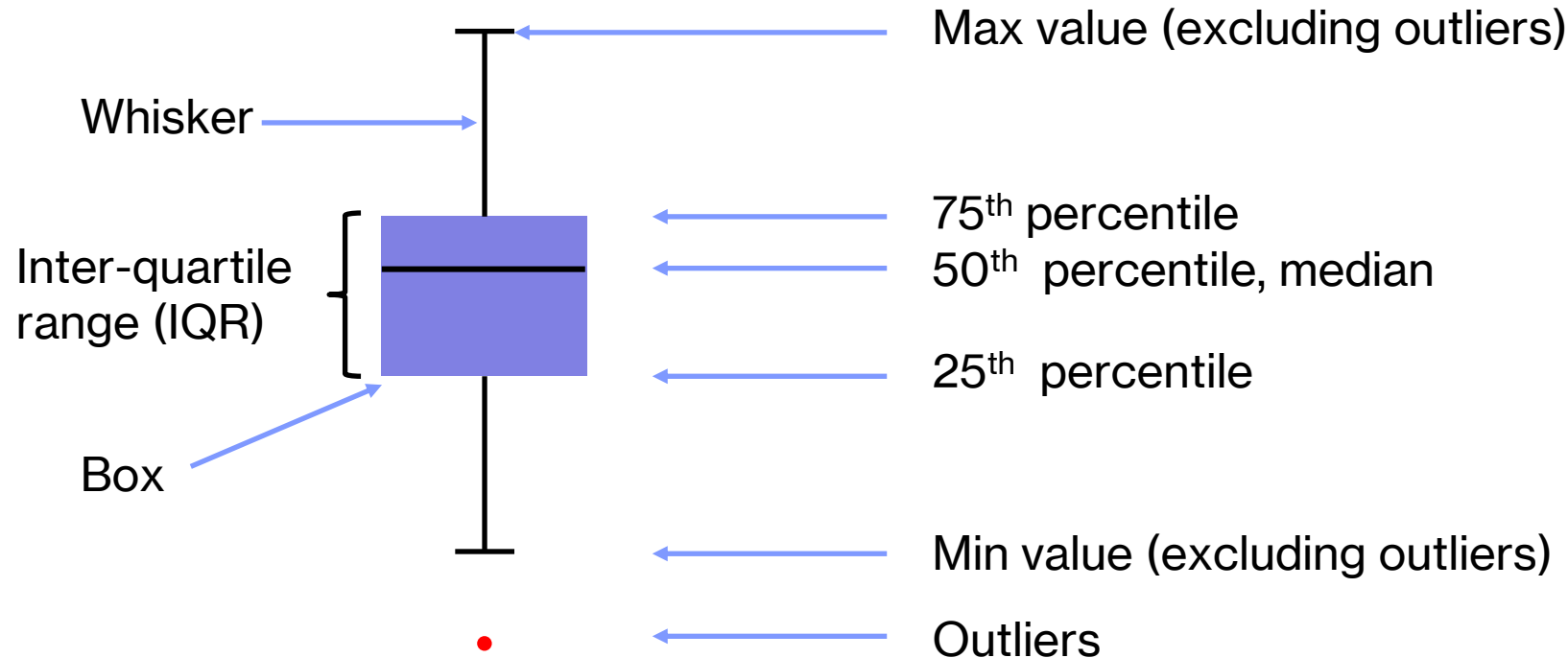
- They ‘suck’ (technical term)
- They do a bad job of communicating the data
- There are much better alternatives
- Lots of papers trying to make people stop using them!
- “*Show the dots in plots*”
 - Nature Biomedical engineering
- “*Beyond Bar and Line Graphs: Time for a New Data Presentation Paradigm*”
 - Weissgerber et al. 2015, PLOS Biology
- “*Reveal, Don’t Conceal*”
 - Weissgerber et al. 2019, Circulation research
- On Blackboard



Hint/Rule 5:
do not use bar charts

Boxplots or 'Box and Whisker' Plots

- Multiple styles available



- MATLAB**: By default, an outlier is a value that is more than 1.5 times the interquartile range away from the bottom or top of the box
 - Other outlier specifications covered later

Hint/Rule 6
define your
whiskers and
outliers

Hint/Rule 7
consider
whether the
box or
whiskers add
understanding

Creating a boxplot/dotplot mashup

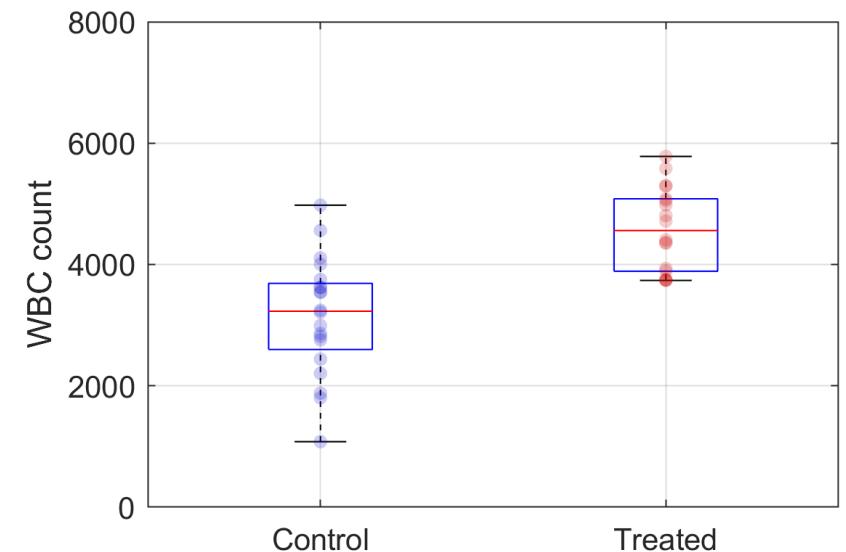
% Boxplot

```
subplot(153);hold on;grid on; box on;title('Box plot')
Data=[Control; Treated];           % creates a 1D array
Groups=[zeros(NC,1); ones(NT,1)]; % creates a 1D array
boxplot(Data,Groups);               % Creates the boxplot

axis([0.5 2.5 0 8000])              % sets x limits (1,2)
set(gca,'xticklabel',{'Control','Treated'}) % set x axis labels
ylabel('WBC count')                 % set y axis label
```

% Make it better by showing the Dots in Plots

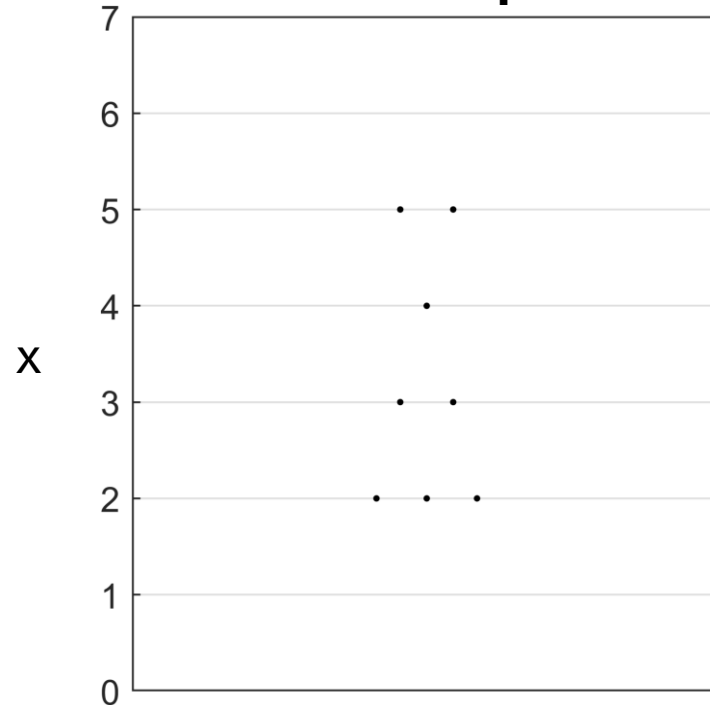
```
Size=20;
DC=scatter(1,Control,Size,Col{1},'filled');
DT=scatter(2,Treated,Size,Col{2},'filled');
alpha(DC,0.2) % Make markers transparent
alpha(DT,0.2)
```



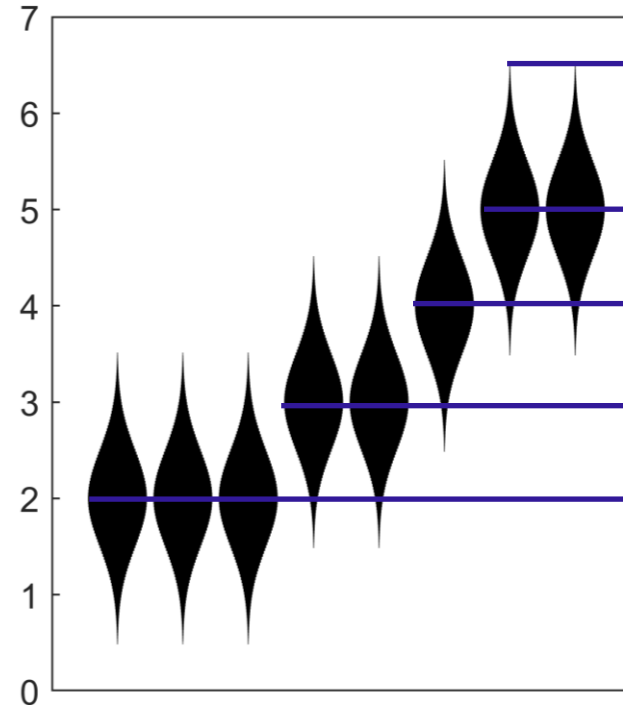
Beeswarm + Violin Plots

```
x=[2 2 2 3 3 4 5 5]; % generate an arbitrary array
```

Beeswarm plot

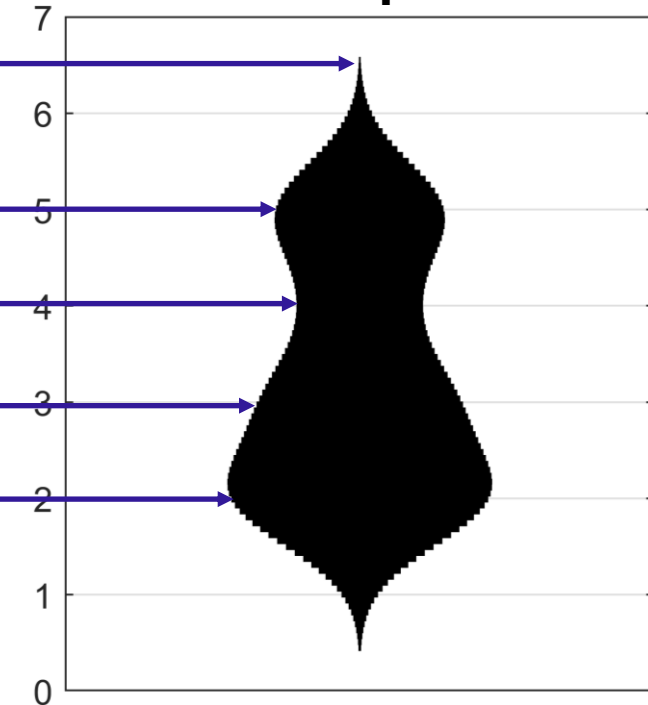


Beeswarm plot scatters values along x axis to help with visualisation



Imagine now that each data point is represented by a normal distribution shaped 'kernel'

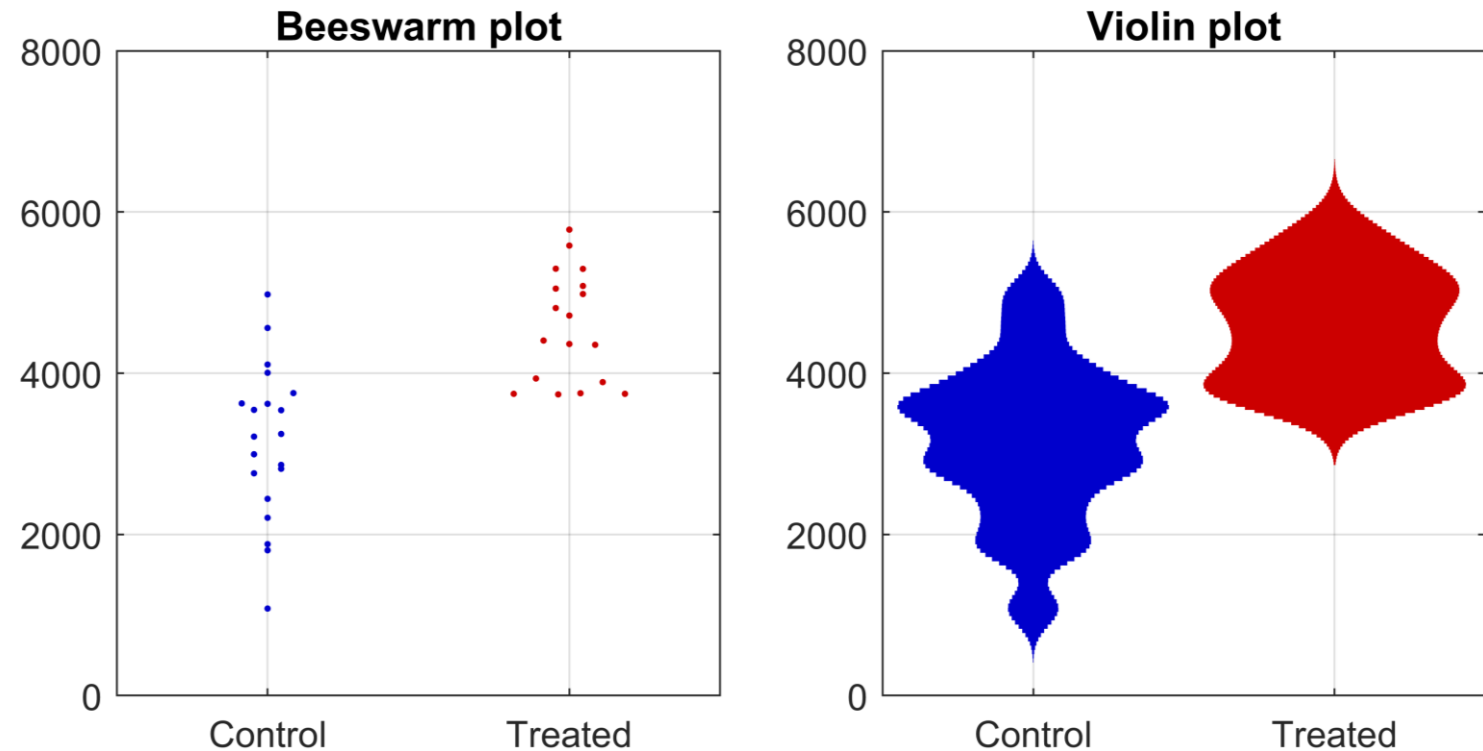
Violin plot



For each location on the x axis, sum up the total black pixels. This is a violin plot

Better plots for WBC count

- Beeswarm plot generated using `plotSpread`
- Violin plot generated using `distributionPlot`
- Both functions from MATLAB central (and on Blackboard)
- Try in the exercises



Hint/Rule 8

consider whether the violin actually obscures the true nature of the data

Descriptive Statistics and Visualisation

What we covered

- Course Practicalities
- What are probability and statistics?
- Course overview
- Descriptive Statistics
 - Population vs Sample
 - Parametric vs Non-parametric
 - Mean, Median, Percentiles,
 - Median absolute deviation, standard deviation, variance
- Data visualisation
 - Dot plots, Histograms, Boxplots, Beeswarm plots, Violin plots

Next Up

Lecture 2: Probability Theory

